# ABSTRACT

The rapid escalation of healthcare costs has made it increasingly important to predict medical insurance expenses accurately. This project presents a Medical Insurance Cost Prediction System, which combines statistical analysis, machine learning, and modern web technologies to deliver real-time cost predictions based on user inputs.

The system utilizes a structured dataset containing demographic and behavioral attributes such as age, gender, body mass index (BMI), smoking habits, number of dependents, and geographic region. Extensive exploratory data analysis (EDA) was conducted using Pandas, Seaborn, and Matplotlib to identify patterns, detect outliers, and understand feature relationships.

A Multiple Linear Regression model was implemented using Scikit-learn. The model achieved an $R^2$ score of approximately 0.75, demonstrating strong predictive capability while maintaining interpretability.

The trained model was deployed via a RESTful API using Flask, and a user-friendly frontend was developed using Next.js with Tailwind CSS. The application supports region mapping, currency conversion, and an informational interface on insurance providers.

This project demonstrates how machine learning can be effectively integrated into a full-stack system to provide practical, scalable, and user-centric solutions in the healthcare domain..

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# CHAPTER 1

# INTRODUCTION

**Background**

The healthcare industry generates vast amounts of data, making it an ideal domain for applying machine learning techniques. Insurance providers rely on predictive models to estimate risk and determine premiums. However, these models are often opaque and inaccessible to end users.

This project bridges the gap by providing a transparent, interactive system that allows users to estimate their medical insurance costs based on personal attributes.

**Problem Statement**

Medical insurance costs vary widely due to multiple interacting factors. Traditional estimation methods are either manual, inaccurate, or inaccessible. There is a need for a system that:

- Accurately predicts insurance costs
- Provides real-time user interaction
- Is interpretable and scalable

**Objectives**

- Perform statistical analysis to understand insurance cost drivers
- Develop a robust regression-based predictive model
- Deploy the model through a scalable backend API
- Build an intuitive and responsive frontend interface
- Enhance usability through additional features like currency conversion

**Scope of the Project**

The system focuses on:

- Predictive modeling using structured tabular data
- Web-based interaction for real-time predictions
- Deployment-ready architecture

**Limitations include:**

- Use of a relatively small dataset
- Linear model assumptions
- No deep learning or real-time insurance data integration

# CHAPTER 2
# IMPLEMENTATION

## 1. Data Analysis and Preprocessing

### Dataset Overview

The dataset consists of 7 attributes:

| Feature | Type | Description |
|---|---|---|
| Age | Numerical | Age of the individual |
| Sex | Categorical | Gender |
| BMI | Numerical | Body Mass Index |
| Children | Numerical | Number of dependents |
| Smoker | Categorical | Smoking status |
| Region | Categorical | Residential region |
| Charges | Numerical | Insurance cost |

### Exploratory Data Analysis (EDA)

EDA revealed several important insights:

- Right-skewed distribution of insurance charges
- Strong correlation between smoking and high charges
- Moderate correlation between age and charges
- Presence of outliers in BMI and charges

### Visualization tools used:

- Histograms → distribution analysis
- Boxplots → outlier detection
- Scatterplots → feature relationships

**Feature Engineering**

- Categorical encoding using label encoding
- No missing values detected
- Feature scaling not required for linear regression

## 2. Machine Learning Model

**Mathematical Model**

The Multiple Linear Regression model is defined as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

Where:

- $y$= predicted insurance cost
- $x_i$= independent variables
- $\beta_i$= coefficients
- $\epsilon$= error term

**Training Process**

- Dataset split: 80% training, 20% testing
- Model trained using Ordinary Least Squares (OLS)
- Implemented using Scikit-learn

**Model Evaluation Metrics**

- **$R^2$ Score:** ~0.75
- **Mean Absolute Error (MAE):** Measures average prediction error
- **Mean Squared Error (MSE):** Penalizes larger errors

**Interpretability**

The model provides insight into feature importance:

- Smoking status → highest impact
- BMI → moderate impact
- Age → gradual increase effect

**3. Backend API (Flask)**

The backend system was implemented using Flask.

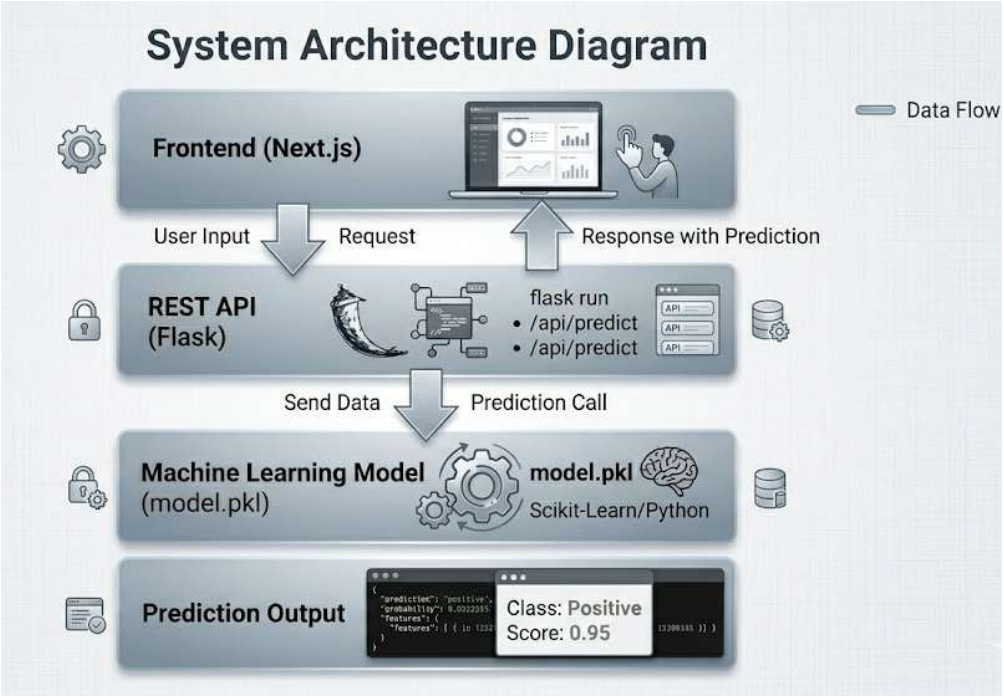



Fig 2.1: System Architecture

**Key Functionalities**

- Input validation
- Data preprocessing
- Model inference
- JSON response handling
- Cross-Origin Resource Sharing (CORS)

**Error Handling**

- Invalid inputs handled gracefully
- HTTP status codes used for communication
- Logging for debugging

**4. Frontend Application**

The frontend was built using Next.js and styled with Tailwind CSS.

**User Interface Design**

- Glassmorphism UI design
- Dark mode support
- Responsive layout

**Core Features**

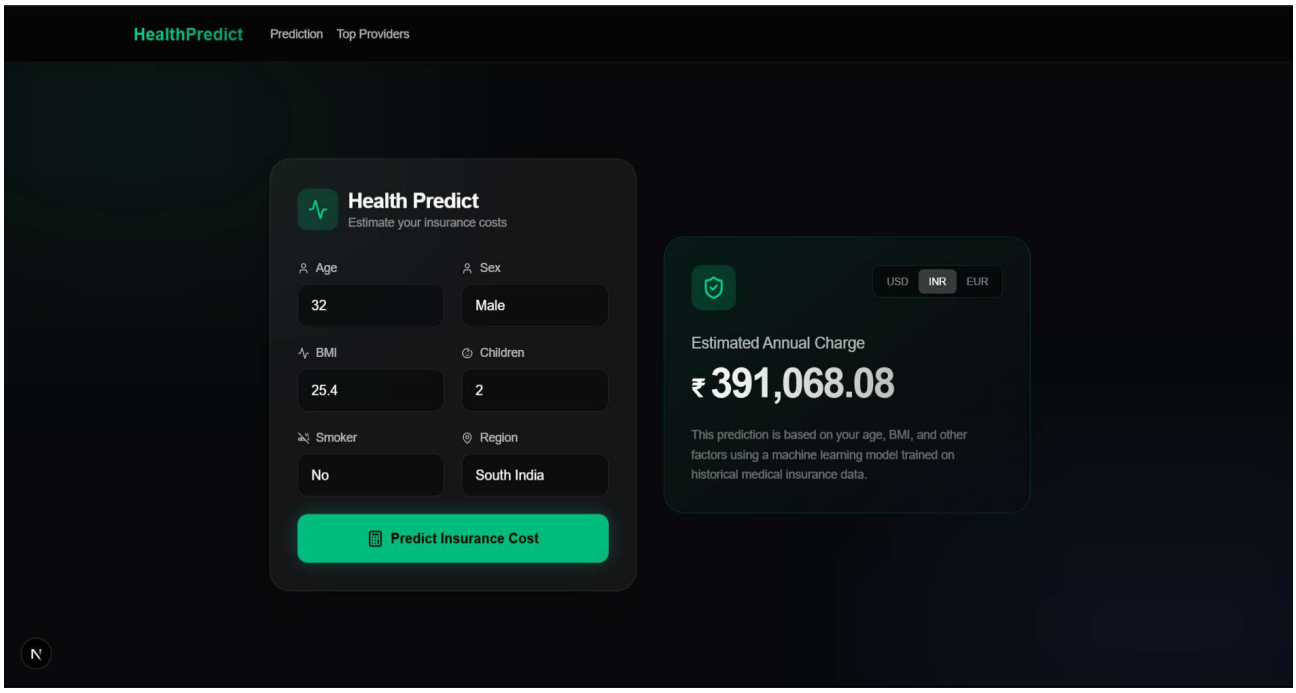
1. **Prediction Form**
   User inputs health and demographic data
2. **Real-Time Prediction**
   Displays predicted insurance cost instantly
3. **Currency Converter**
   Converts output into INR, USD, EUR
4. **Region Mapping**
   Maps Indian regions to dataset categories
5. **Top Providers Page**

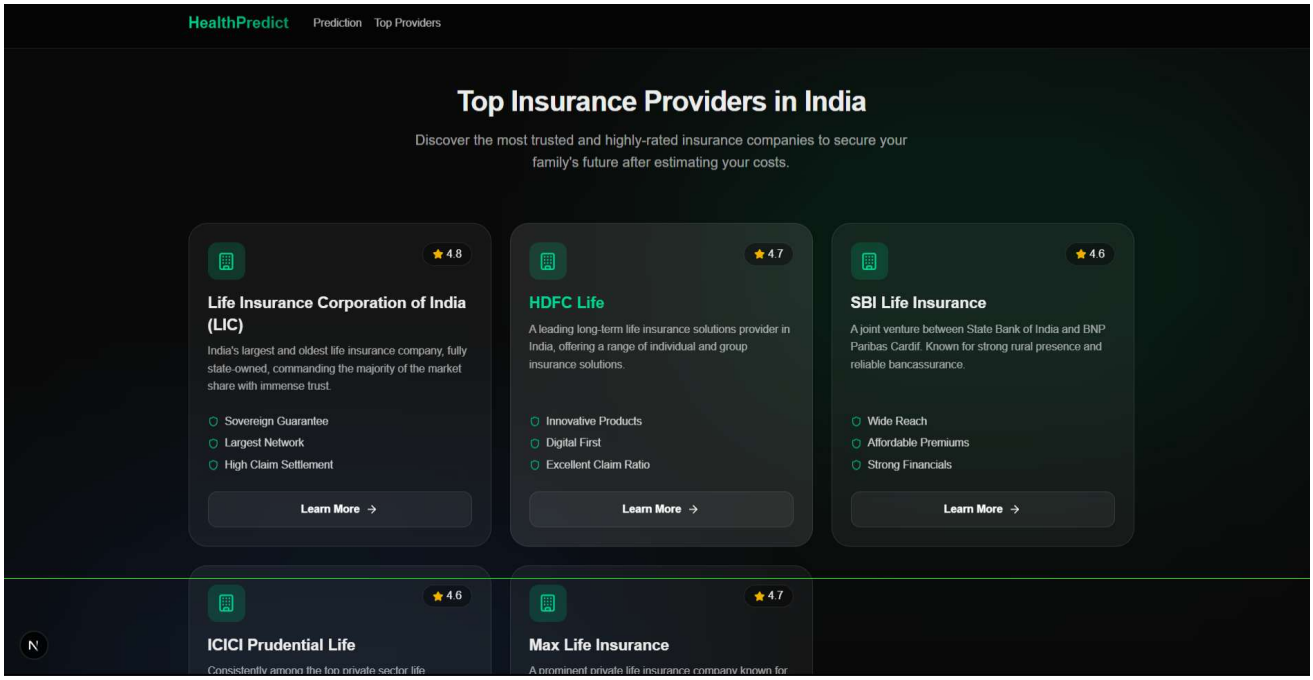
**Frontend-Backend Communication**

- API calls using fetch/axios
- JSON data exchange
- Asynchronous request handling

# CHAPTER 3

# RESULTS



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